FURTHER MECHANICS / WORK, ENERGY AND POWER

Elastic Potential Energy

Worked solutions 10 questions · 101 marks

Instructions

Unless a question states otherwise, take $g = 9.8\,\mathrm{m\,s^{-2}}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10\,\mathrm{m\,s^{-2}}$ where these differ.

Questions

Question 1	*6 marks*	2
Question 2	*11 marks*	4
Question 3	*6 marks*	6
Question 4	*9 marks*	8
Question 5	*13 marks*	11
Question 6	*11 marks*	14
Question 7	*12 marks*	17
Question 8	*11 marks*	19
Question 9	*11 marks*	22
Question 10	*11 marks*	25

1. One end of a light elastic string, of natural length $2a$ and modulus of elasticity $2mg$, is attached to a fixed point O. The other end of the string is attached to a particle P of mass m. The string is vertical with P directly above O. The particle P is pulled vertically upwards until the extension of the string is $2a$. The particle P is then released from rest.

(a) Find the speed of P when it is at a distance $\frac{3}{2}a$ above O. **[4]**

(b) Find the initial acceleration of P when it is released from rest. **[2]**

Solution

(a) The natural length of the string is $2a$.

Initially the extension is $2a$, so the initial length is

$$2a + 2a = 4a$$

Hence initially P is $4a$ above O.

When P is $\frac{3}{2}a$ above O, the length of the string is $\frac{3}{2}a < 2a$, so the string is slack. Therefore at this point the elastic potential energy is 0.

The work–energy principle is

$$KE_0 + GPE_0 + EPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1 + EPE_1$$

where state 0 is the moment of release and state 1 the moment when P is $\frac{3}{2}a$ above O. No other force does work, so $\sum W_{\text{other}} = 0$.

Initially, P is released from rest, so $KE_0 = 0$.

The elastic potential energy is

$$EPE = \frac{\lambda x^2}{2l}$$

where x is the extension and l the natural length. Here $\lambda = 2mg$ and $l = 2a$, and initially $x = 2a$, so

$$EPE_0 = \frac{(2mg)(2a)^2}{2(2a)} = 2mga$$

Take the position in state 1 as the zero level for gravitational potential energy. From state 0 to state 1, P falls $4a - \frac{3}{2}a = \frac{5}{2}a$, so

$$GPE_0 = mg\left(\frac{5a}{2}\right) = \frac{5}{2}mga, \qquad GPE_1 = 0$$

Also

$$KE_1 = \frac{1}{2}mv^2, \qquad EPE_1 = 0$$

So

$$0 + \frac{5}{2}mga + 2mga + 0 = \frac{1}{2}mv^2 + 0 + 0$$

$$\frac{1}{2}mv^2 = \frac{9}{2}mga$$

Hence

$$v^2 = 9ag$$

so

$$v = 3\sqrt{ag}$$

(b) The tension is

$$T = \frac{\lambda x}{l}$$

where x is the extension and l the natural length. Initially the extension is $2a$, so

$$T = \frac{(2mg)(2a)}{2a} = 2mg$$

This tension acts downwards, towards O. The weight mg also acts downwards.

So the resultant force on P initially is

$$2mg + mg = 3mg$$

downwards.

Taking downward as positive and denoting the acceleration by α, Newton's second law gives

$$m\alpha = 3mg$$

so

$$\alpha = 3g$$

Therefore the initial acceleration is $3g$ downwards.

2. A light elastic string has natural length 2.5 m and modulus of elasticity 15 N.

One end of the elastic string is attached to a particle of mass 0.75 kg.

The other end of the elastic string is attached to a fixed point O.

The particle is released from rest at a point A, which is 2.9 m vertically below O.

(a) Calculate the elastic potential energy stored in the string when the particle is at A. **[2]**

(b) The point B is 4.0 m vertically below O.
Calculate the gravitational potential energy lost by the particle as it moves from A to B. **[2]**

(c) Find the speed of the particle at B. **[4]**

(d) The point C is 3.3 m vertically below O.
Explain, showing any calculations that you make, why the speed of the particle is increasing the first time that the particle is at C. **[3]**

Solution

(a) The extension of the string at A is

$$x_A = 2.9 - 2.5 = 0.4\,\text{m}$$

The elastic potential energy is

$$EPE = \frac{\lambda x^2}{2l}$$

where x is the extension and l the natural length. So

$$EPE_A = \frac{15(0.4)^2}{2(2.5)} = 0.48\,\text{J}$$

(b) The particle falls from 2.9 m below O to 4.0 m below O, so the vertical distance moved down is

$$4.0 - 2.9 = 1.1\,\text{m}$$

Therefore the gravitational potential energy lost is

$$mg \times 1.1 = 0.75 \times 9.8 \times 1.1 = 8.085\,\text{J}$$

(c) At B, the extension is

$$x_B = 4.0 - 2.5 = 1.5\,\text{m}$$

So the elastic potential energy at B is

$$EPE_B = \frac{15(1.5)^2}{2(2.5)} = 6.75\,\text{J}$$

The work–energy principle is

$$KE_0 + GPE_0 + EPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1 + EPE_1$$

where state 0 is the particle at A and state 1 the particle at B. No other force does work, so $\sum W_{\text{other}} = 0$. The particle is released from rest, so $KE_A = 0$. Also, from part (b), the loss in GPE from A to B is $GPE_A - GPE_B = 8.085\,\text{J}$. Therefore

$$KE_B = EPE_A + (GPE_A - GPE_B) - EPE_B$$

$$KE_B = 0.48 + 8.085 - 6.75 = 1.815\,\text{J}$$

Now

$$\frac{1}{2} \times 0.75 \times v^2 = 1.815$$

so

$$v^2 = \frac{2 \times 1.815}{0.75} = 4.84$$

$$v = 2.2\,\text{m}\,\text{s}^{-1}$$

(d) At C, the extension is

$$x_C = 3.3 - 2.5 = 0.8\,\text{m}$$

The tension is

$$T = \frac{\lambda x}{l}$$

where x is the extension and l the natural length. So the tension in the string is

$$T = \frac{15 \times 0.8}{2.5} = 4.8\,\text{N}$$

This acts upwards.

The weight of the particle is

$$0.75 \times 9.8 = 7.35\,\text{N}$$

This acts downwards.

Hence the resultant force is

$$7.35 - 4.8 = 2.55\,\text{N}$$

downwards, so the acceleration is downwards.

The first time the particle is at C, it is moving downwards because it has fallen from A towards B. Since the acceleration is also downwards, the acceleration is in the same direction as the motion.

Therefore the speed is increasing the first time that the particle is at C.

Final answers with $g = 10$: (b) 8.25 J; (c) $2.30\,\text{m}\,\text{s}^{-1}$

3. A particle P of mass m is attached to one end of a light elastic string of natural length a and modulus $2mg$. The other end of the string is attached to a fixed point O on a rough plane inclined at $30°$ to the horizontal.

The string lies along a line of greatest slope, with O below P. The particle is initially at rest on the plane with the string at its natural length.

The coefficient of friction between P and the plane is $\dfrac{1}{\sqrt{3}}$.

The particle is projected up the plane, away from O, with speed $\sqrt{\dfrac{3ga}{2}}$.

Find the greatest extension of the string during the subsequent motion. **[6]**

Solution

Let x be the greatest extension of the string.

While the particle moves up the plane to this point, it is moving upwards, so friction acts down the plane. Since the string lies along the plane, it has no component perpendicular to the plane, so

$$R = mg\cos 30° = \frac{\sqrt{3}}{2}mg$$

Hence the friction force is

$$\mu R = \frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{2}mg = \frac{1}{2}mg$$

Take the initial position as the zero level for gravitational potential energy.

The work–energy principle is

$$KE_0 + GPE_0 + EPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1 + EPE_1$$

where state 0 is the moment of projection and state 1 the moment of greatest extension. The kinetic energies are

$$KE_0 = \frac{1}{2}m\left(\sqrt{\frac{3ga}{2}}\right)^2 = \frac{3mga}{4}, \qquad KE_1 = 0$$

Also, after moving a distance x up the plane, the vertical rise is $x\sin 30° = \frac{x}{2}$, so

$$GPE_0 = 0, \qquad GPE_1 = mg\left(\frac{x}{2}\right) = \frac{1}{2}mgx$$

The elastic potential energy is

$$EPE = \frac{\lambda x^2}{2l}$$

where x is the extension and l the natural length. Here $l = a$, and the string starts at natural length, so

$$EPE_0 = 0, \qquad EPE_1 = \frac{(2mg)x^2}{2a} = \frac{mgx^2}{a}$$

The only other force doing work is friction, which opposes the motion, so

$$\sum W_{\text{other}} = -\left(\frac{1}{2}mg\right)x = -\frac{1}{2}mgx$$

Substituting into the energy equation:

$$\frac{3mga}{4} - \frac{1}{2}mgx = \frac{1}{2}mgx + \frac{mgx^2}{a}$$

So

$$\frac{3mga}{4} = mgx + \frac{mgx^2}{a}$$

Divide through by mg:

$$\frac{3a}{4} = x + \frac{x^2}{a}$$

Multiply through by a:

$$\frac{3a^2}{4} = ax + x^2$$

Hence

$$x^2 + ax - \frac{3a^2}{4} = 0$$

Solving this quadratic:

$$x = \frac{-a \pm \sqrt{a^2 + 3a^2}}{2} = \frac{-a \pm 2a}{2}$$

So

$$x = \frac{a}{2} \quad \text{or} \quad x = -\frac{3a}{2}$$

Since x is an extension, it must be positive. Therefore

$$x = \frac{a}{2}$$

4. The points A and B are at the same horizontal level and are $24a$ apart. The ends of a light elastic string, of natural length $18a$ and modulus of elasticity λ, are attached to A and B. A particle P of mass m is attached to the midpoint of the string. The system is in equilibrium with P at a distance $5a$ below M, the midpoint of AB.

(a) Find λ in terms of m and g. **[3]**

The particle P is pulled down vertically and released from rest from a position $9a$ below M.

(b) Find, in terms of a and g, the speed of P as it passes through its equilibrium position. **[6]**

Solution

(a) At equilibrium:

$$AM = 12a, \qquad MP = 5a$$

so

$$AP = \sqrt{(12a)^2 + (5a)^2} = 13a$$

Hence the actual length of the whole string is

$$2 \times 13a = 26a$$

so its extension is

$$26a - 18a = 8a$$

The tension is

$$T = \frac{\lambda x}{l}$$

where x is the extension and l the natural length. The tension T is the same in each side, and here $x = 8a$ and $l = 18a$, so

$$T = \frac{\lambda(8a)}{18a} = \frac{4\lambda}{9}$$

If θ is the angle each side makes with the horizontal, then from the 5-12-13 triangle:

$$\sin\theta = \frac{5}{13}$$

Resolving vertically for equilibrium:

$$2T\sin\theta = mg$$

$$2\left(\frac{4\lambda}{9}\right)\left(\frac{5}{13}\right) = mg$$

$$\frac{40\lambda}{117} = mg$$

Therefore

$$\lambda = \frac{117mg}{40}$$

(b) When P is $9a$ below M:

$$AP = \sqrt{(12a)^2 + (9a)^2} = 15a$$

So the whole string has length

$$2 \times 15a = 30a$$

and therefore its extension is

$$30a - 18a = 12a$$

The elastic potential energy is

$$EPE = \frac{\lambda x^2}{2l}$$

where x is the extension and l the natural length.

Initially:

$$EPE_0 = \frac{\lambda(12a)^2}{2(18a)}$$

$$= 4\lambda a$$

At the equilibrium position, the extension is $8a$, so

$$EPE_1 = \frac{\lambda(8a)^2}{2(18a)}$$

$$= \frac{16\lambda a}{9}$$

From the release point to equilibrium, P rises by

$$9a - 5a = 4a$$

Take the release point as the zero level for gravitational potential energy. Then

$$GPE_0 = 0, \qquad GPE_1 = mg(4a) = 4mga$$

The work–energy principle is

$$KE_0 + GPE_0 + EPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1 + EPE_1$$

where state 0 is the moment of release and state 1 the moment P passes through its equilibrium position. No other force does work, so $\sum W_{\text{other}} = 0$. The particle is released from rest, so $KE_0 = 0$, and

$$0 + 0 + 4\lambda a + 0 = \frac{1}{2}mv^2 + 4mga + \frac{16\lambda a}{9}$$

$$\frac{20\lambda a}{9} = 4mga + \frac{1}{2}mv^2$$

Substitute $\lambda = \frac{117mg}{40}$:

$$\frac{20a}{9} \times \frac{117mg}{40} = 4mga + \frac{1}{2}mv^2$$

$$\frac{13}{2}mga = 4mga + \frac{1}{2}mv^2$$

$$\frac{5}{2}mga = \frac{1}{2}mv^2$$

$$v^2 = 5ag$$

Therefore

$$v = \sqrt{5ag}$$

5. A light elastic string has natural length $3a$ and modulus of elasticity $9mg$.
One end of the string is attached to a fixed point O. A particle P of mass m is attached to the other end.
The particle hangs freely in equilibrium at the point E, vertically below O.

(a) Find the length OE. **[3]**

The particle is then pulled vertically downwards from E to a point A, where $OA = \frac{16a}{3}$, and released from rest.
The resistance to the motion of P is a constant force of magnitude $\frac{1}{4}mg$.

(b) Find, in terms of a and g, the speed of P as it passes through E. **[6]**

The particle is now held at O and released from rest.
The resistance to the motion of P is again a constant force of magnitude $\frac{1}{4}mg$.
The speed of P is greatest when it is at the point B.

(c) Find the distance OB. **[4]**

Solution

(a) Let the extension of the string in equilibrium be x.

The tension is

$$T = \frac{\lambda x}{l}$$

where x is the extension and l the natural length. Here $l = 3a$, so

$$T = \frac{9mg\,x}{3a} = \frac{3mgx}{a}$$

At equilibrium, the tension balances the weight, so

$$\frac{3mgx}{a} = mg$$

$$x = \frac{a}{3}$$

Hence the length of the string is

$$OE = 3a + \frac{a}{3} = \frac{10a}{3}$$

(b) From part (a), the extension at E is

$$OE - 3a = \frac{10a}{3} - 3a = \frac{a}{3}$$

At A:

$$OA = \frac{16a}{3}$$

so the extension at A is

$$\frac{16a}{3} - 3a = \frac{7a}{3}$$

Also

$$AE = \frac{16a}{3} - \frac{10a}{3} = 2a$$

The elastic potential energy is

$$EPE = \frac{\lambda x^2}{2l}$$

where x is the extension and l the natural length. Here $l = 3a$.

So at A:

$$EPE_A = \frac{9mg}{2(3a)}\left(\frac{7a}{3}\right)^2 = \frac{49}{6}mga$$

and at E:

$$EPE_E = \frac{9mg}{2(3a)}\left(\frac{a}{3}\right)^2 = \frac{1}{6}mga$$

Take A as the zero level for gravitational potential energy. Then

$$GPE_A = 0, \qquad GPE_E = mg(2a) = 2mga$$

The work–energy principle is

$$KE_0 + GPE_0 + EPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1 + EPE_1$$

where state 0 is the particle at A and state 1 the particle at E. The only other force doing work is the resistance, of magnitude $\frac{1}{4}mg$, which acts opposite to the motion over the distance $AE = 2a$, so

$$\sum W_{\text{other}} = -\frac{1}{4}mg(2a) = -\frac{1}{2}mga$$

The particle is released from rest at A, so

$$0 + 0 + \frac{49}{6}mga - \frac{1}{2}mga = \frac{1}{2}mv^2 + 2mga + \frac{1}{6}mga$$

So

$$\frac{46}{6}mga = \frac{1}{2}mv^2 + \frac{13}{6}mga$$

$$\frac{1}{2}mv^2 = \frac{33}{6}mga = \frac{11}{2}mga$$

$$v^2 = 11ag$$

Hence the speed as P passes through E is

$$v = \sqrt{11ag}$$

(c) The particle is released from rest at O.

While the string is slack, there is no tension. The forces are weight mg downwards and resistance $\frac{1}{4}mg$ upwards, so the resultant force is

$$mg - \frac{1}{4}mg = \frac{3}{4}mg$$

downwards.

So at first the particle accelerates downwards and its speed increases.

Once the string becomes taut, let the extension be x. Then the tension is

$$T = \frac{\lambda x}{3a} = \frac{9mg\,x}{3a} = \frac{3mgx}{a}$$

The downward resultant force is now

$$mg - \frac{1}{4}mg - T = \frac{3}{4}mg - \frac{3mgx}{a}$$

The speed is greatest when the acceleration is zero, so the resultant force is zero:

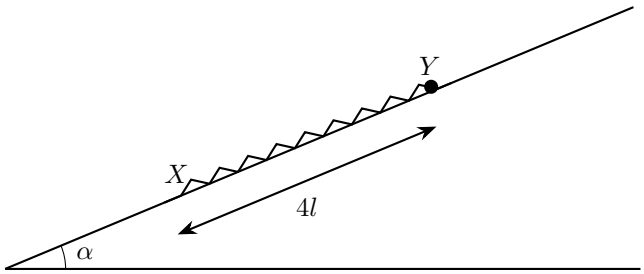
$$\frac{3}{4}mg - \frac{3mgx}{a} = 0$$

$$\frac{3}{4} = \frac{3x}{a}$$

$$x = \frac{a}{4}$$

So the extension at B is $\frac{a}{4}$, and

$$OB = 3a + \frac{a}{4} = \frac{13a}{4}$$



6. A light elastic spring has natural length $2l$ and modulus of elasticity mg. One end of the spring is attached to a fixed point X on a rough inclined plane. The other end is attached to a package P of mass m.

The plane is inclined to the horizontal at an angle α where

$$\tan\alpha = \frac{5}{12}$$

The package is initially held at the point Y on the plane, where $XY = 4l$. The point Y is higher than X and XY is a line of greatest slope of the plane, as shown. The package is released from rest at Y and moves down the plane.

The coefficient of friction between P and the plane is $\frac{1}{4}$.

By modelling P as a particle,

(a) show that the acceleration of P at the instant when it is released is

$$\frac{15}{13}g$$

down the plane. **[5]**

(b) find, in terms of g and l, the speed of P at the instant when the spring first regains its natural length. **[6]**

Solution

(a) Since

$$\tan\alpha = \frac{5}{12}$$

we have

$$\sin\alpha = \frac{5}{13}, \qquad \cos\alpha = \frac{12}{13}$$

Initially the spring has length $4l$ and natural length $2l$, so its extension is

$$4l - 2l = 2l$$

The tension is

$$T = \frac{\lambda x}{L}$$

where x is the extension and L the natural length. Here $\lambda = mg$, $x = 2l$ and $L = 2l$, so

$$T = \frac{mg(2l)}{2l} = mg$$

This tension acts down the plane, towards X.

Since the spring lies along the plane, it has no component perpendicular to the plane. Hence the normal reaction is

$$R = mg\cos\alpha = mg \times \frac{12}{13} = \frac{12mg}{13}$$

So the friction is

$$F = \mu R = \frac{1}{4} \times \frac{12mg}{13} = \frac{3mg}{13}$$

and this acts up the plane.

The component of the weight down the plane is

$$mg\sin\alpha = mg \times \frac{5}{13} = \frac{5mg}{13}$$

Therefore the resultant force down the plane is

$$\begin{aligned} T + mg\sin\alpha - F &= mg + \frac{5mg}{13} - \frac{3mg}{13} \\ &= mg + \frac{2mg}{13} \\ &= \frac{15mg}{13} \end{aligned}$$

Using Newton's second law:

$$ma = \frac{15mg}{13}$$

so

$$a = \frac{15}{13}g$$

Therefore the acceleration is $\frac{15}{13}g$ down the plane.

(b) Let the instant of release be state 0, and let state 1 be the instant when the spring first regains its natural length.

The spring length changes from $4l$ to $2l$, so the package moves

$$4l - 2l = 2l$$

down the plane.

The work–energy principle is

$$KE_0 + GPE_0 + EPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1 + EPE_1$$

Initially the package is at rest, so

$$KE_0 = 0$$

At the instant when the spring regains its natural length:

$$EPE_1 = 0$$

The elastic potential energy is

$$EPE = \frac{\lambda x^2}{2L}$$

where x is the extension and L the natural length. Here $L = 2l$, and initially $x = 2l$, so

$$EPE_0 = \frac{mg(2l)^2}{2(2l)} = mgl$$

The vertical drop of the package is

$$2l\sin\alpha = 2l \times \frac{5}{13} = \frac{10l}{13}$$

Take the position in state 1 as the zero level for gravitational potential energy. Then

$$GPE_0 = mg \times \frac{10l}{13} = \frac{10mgl}{13}$$

The only other force doing work is friction, of constant magnitude $\frac{3mg}{13}$, which opposes the motion over the distance $2l$, so

$$\sum W_{\text{other}} = -\frac{3mg}{13} \times 2l = -\frac{6mgl}{13}$$

Substitute into the energy equation:

$$0 + \frac{10mgl}{13} + mgl - \frac{6mgl}{13} = \frac{1}{2}mv^2 + 0 + 0$$

$$mgl + \frac{4mgl}{13} = \frac{1}{2}mv^2$$

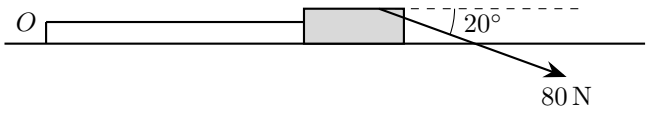
$$\frac{17mgl}{13} = \frac{1}{2}mv^2$$

Therefore

$$v^2 = \frac{34gl}{13}$$

and so

$$v = \sqrt{\frac{34gl}{13}}$$



7. A block of mass 6 kg rests on a rough horizontal surface and is attached to a fixed point O by a light elastic string. Initially the block is at rest 2.4 m from O.

The elastic string has natural length 2 m and modulus of elasticity 80 N.

The coefficient of friction between the block and the surface is 0.25.
A force of magnitude 80 N is applied to the block at an angle of 20° below the horizontal, with its horizontal component directed away from O, as shown in the diagram.

The block moves in a straight line directly away from O.
In this question, take $g = 9.8\,\text{m}\,\text{s}^{-2}$.

(a) Show that the block has moved 0.94 m, correct to 2 significant figures, when the forces acting on it are in equilibrium. **[4]**

(b) State one limitation of the model. **[1]**

(c) Find the maximum speed of the block. **[7]**

Solution

(a) Let the block move a distance x m away from O.

The string length is then $2.4 + x$, so its extension is

$$(2.4 + x) - 2 = 0.4 + x$$

The tension is

$$T = \frac{\lambda e}{l}$$

where e is the extension and l the natural length. Here $e = 0.4 + x$, so

$$T = \frac{80(0.4 + x)}{2} = 16 + 40x$$

Since the block moves horizontally, it is in vertical equilibrium, so

$$R = 6g + 80\sin 20^\circ = 58.8 + 80\sin 20^\circ$$

Therefore the friction is

$$F = \mu R = 0.25(58.8 + 80\sin 20^\circ) = 21.54\,\text{N}$$

When the forces are in equilibrium, the horizontal forces balance:

$$80\cos 20^\circ = T + F$$

$$80\cos 20^\circ = (16 + 40x) + 21.54$$

So

$$40x = 80\cos 20^\circ - 37.54$$

$$x \approx 0.9409$$

Hence the block has moved

$$0.94\,\text{m} \qquad (2\text{ s.f.})$$

(b) One limitation is that the coefficient of friction is assumed to remain constant throughout the motion.

(c) The speed is greatest where the resultant force is zero, at the equilibrium position. From part (a), this is at $x = 0.9409\,\text{m}$.

The elastic potential energy is

$$EPE = \frac{\lambda e^2}{2l}$$

where e is the extension and l the natural length. Initially the extension is $0.4\,\text{m}$, so

$$EPE_0 = \frac{80(0.4)^2}{2 \times 2} = 3.2\,\text{J}$$

At the position of maximum speed, the extension is

$$e_1 = 0.4 + 0.9409 = 1.3409$$

The work–energy principle is

$$KE_0 + GPE_0 + EPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1 + EPE_1$$

where state 0 is the initial position and state 1 the position of maximum speed. There is no change in gravitational potential energy. The other forces doing work are the applied force and friction, which opposes the motion, so

$$\sum W_{\text{other}} = 80\cos 20^\circ \times 0.9409 - 21.54 \times 0.9409$$

Therefore

$$0 + 3.2 + 80\cos 20^\circ(0.9409) - 21.54(0.9409) = \frac{1}{2}(6)v^2 + \frac{80(1.3409)^2}{2 \times 2}$$

$$53.66 = 3v^2 + 35.96$$

$$3v^2 = 17.70$$

$$v = 2.43\,\text{m}\,\text{s}^{-1}$$

8. A spring of natural length a is suspended from a fixed point A. A package P of mass m is attached to the lower end of the spring.

The package is initially at the point B, vertically below A, where $AB = a$.

The package is projected vertically downwards from B with speed $\sqrt{ag}$.

It first comes to instantaneous rest at the point C, where $AC = 2a$.

Air resistance is modelled as negligible. By modelling the spring as light and modelling P as a particle,

(a) show that the modulus of elasticity of the spring is $3mg$. **[4]**

(b) (i) Show that P attains its maximum speed when the extension of the spring is $\frac{1}{3}a$. **[2]**

(ii) Hence find the maximum speed, giving your answer in terms of a and g. **[4]**

(c) In reality, the spring is not light.
State one way in which this would affect your energy equation in part (b). **[1]**

Solution

(a) At B the spring is at its natural length, so its extension is 0.

At C:

$$AC = 2a$$

so the extension is

$$2a - a = a$$

The work–energy principle is

$$KE_0 + GPE_0 + EPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1 + EPE_1$$

where state 0 is P at B and state 1 is P at C. Air resistance is negligible, so no other force does work and $\sum W_{\text{other}} = 0$.

The elastic potential energy is

$$EPE = \frac{\lambda x^2}{2l}$$

where x is the extension and l the natural length. Here $l = a$.

Take C as the zero level for gravitational potential energy.

Then

$$KE_B = \frac{1}{2}m(\sqrt{ag})^2 = \frac{1}{2}mag, \qquad GPE_B = mga, \qquad EPE_B = 0$$

and

$$KE_C = 0, \qquad GPE_C = 0, \qquad EPE_C = \frac{\lambda a^2}{2a} = \frac{\lambda a}{2}$$

So

$$\frac{1}{2}mag + mga = \frac{\lambda a}{2}$$

$$\frac{3}{2}mga = \frac{\lambda a}{2}$$

$$\lambda = 3mg$$

Hence the modulus of elasticity is $3mg$.

(b) **(i)** Let the extension of the spring be x while P is moving downwards.

The tension is

$$T = \frac{\lambda x}{l}$$

where x is the extension and l the natural length. Here $l = a$, so the tension is $\frac{\lambda x}{a}$, acting upwards.

While the particle is moving downwards, its speed is greatest when its acceleration is 0, so the resultant force is 0.

Taking downward as positive:

$$mg - \frac{\lambda x}{a} = 0$$

Using $\lambda = 3mg$:

$$mg - \frac{3mgx}{a} = 0$$

$$1 - \frac{3x}{a} = 0$$

$$x = \frac{a}{3}$$

So P attains its maximum speed when the extension is $\frac{a}{3}$.

(ii) Let the maximum speed be v, and let state 0 be P at B and state 1 be P where the extension is $\frac{a}{3}$.

Take the position in state 1 as the zero level for gravitational potential energy. The particle has fallen a distance $\frac{a}{3}$ from B.

The work–energy principle is

$$KE_0 + GPE_0 + EPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1 + EPE_1$$

No other force does work, so $\sum W_{\text{other}} = 0$, and

$$\frac{1}{2}m(\sqrt{ag})^2 + mg\left(\frac{a}{3}\right) + 0 + 0 = \frac{1}{2}mv^2 + 0 + \frac{\lambda}{2a}\left(\frac{a}{3}\right)^2$$

So

$$\frac{1}{2}mag + \frac{1}{3}mga = \frac{1}{2}mv^2 + \frac{\lambda a}{18}$$

Using $\lambda = 3mg$:

$$\frac{1}{2}mag + \frac{1}{3}mga = \frac{1}{2}mv^2 + \frac{3mga}{18}$$

$$\frac{1}{2}mag + \frac{1}{3}mga = \frac{1}{2}mv^2 + \frac{1}{6}mga$$

$$\frac{1}{2}mv^2 = \left(\frac{1}{2} + \frac{1}{3} - \frac{1}{6}\right)mga = \frac{2}{3}mga$$

$$v^2 = \frac{4ag}{3}$$

Taking the positive root:

$$v = 2\sqrt{\frac{ag}{3}}$$

(c) If the spring is not light, the energy equation would need an extra term for the kinetic energy of the spring.

9. A light elastic string with natural length l and modulus of elasticity kmg has one end attached to a fixed point A on a rough inclined plane. The other end of the string is attached to a package of mass m. The plane is inclined at an angle θ to the horizontal, where

$$\tan\theta = \frac{3}{4}$$

The package is initially held at the point C on a line of greatest slope of the plane below A, where $AC = 3l$. The package is then projected down the plane with speed $2\sqrt{gl}$ and first comes to rest at the point B, where $AB = 5l$.

The coefficient of friction between the package and the plane is $\frac{1}{2}$.

By modelling the package as a particle,

(a) show that

$$k = \frac{2}{5}$$ **[6]**

(b) find the acceleration of the package at the instant it starts to move back up the plane from the point B. **[5]**

Solution

(a) Since

$$\tan\theta = \frac{3}{4}$$

we can use a 3-4-5 triangle, so

$$\sin\theta = \frac{3}{5}, \qquad \cos\theta = \frac{4}{5}$$

Let the modulus of elasticity be

$$\lambda = kmg$$

The package moves from C to B, a distance

$$CB = AB - AC = 5l - 3l = 2l$$

down the plane.

At C, the string length is $3l$, so the extension is

$$x_C = 3l - l = 2l$$

At B, the string length is $5l$, so the extension is

$$x_B = 5l - l = 4l$$

The work–energy principle is

$$KE_0 + GPE_0 + EPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1 + EPE_1$$

where state 0 is the package at C and state 1 the package at B.

The initial kinetic energy is

$$KE_C = \frac{1}{2}m\left(2\sqrt{gl}\right)^2 = 2mgl$$

Since the package moves $2l$ down the plane, the vertical drop is

$$2l\sin\theta = 2l \times \frac{3}{5} = \frac{6l}{5}$$

Take B as the zero level for gravitational potential energy. Then

$$GPE_C = mg \times \frac{6l}{5} = \frac{6mgl}{5}, \qquad GPE_B = 0$$

The friction force has magnitude

$$\mu R = \frac{1}{2}R$$

Since the reaction is

$$R = mg\cos\theta = mg \times \frac{4}{5} = \frac{4mg}{5}$$

the friction force is

$$\frac{1}{2} \times \frac{4mg}{5} = \frac{2mg}{5}$$

The only other force doing work is friction, which opposes the motion over the distance $2l$, so

$$\sum W_{\text{other}} = -\frac{2mg}{5}(2l) = -\frac{4mgl}{5}$$

The elastic potential energy is

$$EPE = \frac{\lambda x^2}{2l}$$

where x is the extension and l the natural length. So at C:

$$EPE_C = \frac{kmg(2l)^2}{2l} = 2kmgl$$

and at B:

$$EPE_B = \frac{kmg(4l)^2}{2l} = 8kmgl$$

The package first comes to rest at B, so

$$KE_B = 0$$

Substitute into the energy equation:

$$2mgl + \frac{6mgl}{5} + 2kmgl - \frac{4mgl}{5} = 0 + 0 + 8kmgl$$

Rearranging:

$$2mgl + \frac{6mgl}{5} - \frac{4mgl}{5} = 8kmgl - 2kmgl$$

So

$$2mgl + \frac{2mgl}{5} = 6kmgl$$

Hence

$$\frac{12mgl}{5} = 6kmgl$$

Dividing by $6mgl$ gives

$$k = \frac{2}{5}$$

(b) The tension is

$$T = \frac{\lambda x}{l}$$

where x is the extension and l the natural length. At B, the extension of the string is $4l$, so

$$T = \frac{kmg \times 4l}{l} = 4kmg$$

Using $k = \frac{2}{5}$:

$$T = 4 \times \frac{2}{5}mg = \frac{8mg}{5}$$

As the package starts to move back up the plane, friction acts down the plane.

The component of weight down the plane is

$$mg\sin\theta = mg \times \frac{3}{5} = \frac{3mg}{5}$$

The normal reaction is

$$R = mg\cos\theta = mg \times \frac{4}{5} = \frac{4mg}{5}$$

So the friction force is

$$\mu R = \frac{1}{2} \times \frac{4mg}{5} = \frac{2mg}{5}$$

Taking up the plane as positive, the resultant force is

$$\frac{8mg}{5} - \frac{3mg}{5} - \frac{2mg}{5} = \frac{3mg}{5}$$

Using Newton's second law:

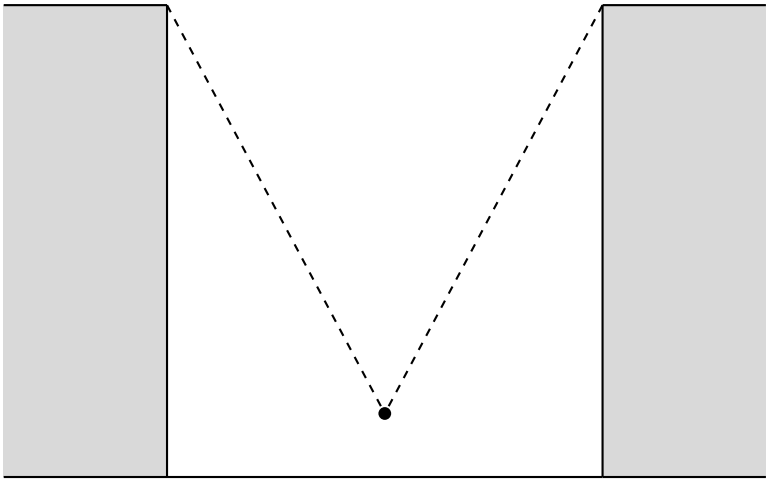
$$ma = \frac{3mg}{5}$$

Therefore

$$a = \frac{3g}{5}$$

So the acceleration is

$$\frac{3g}{5} \text{ up the plane}$$



10. A stunt performer is attached to two identical light elastic cords. One end of each cord is fixed to opposite sides of the rim of a vertical shaft, as shown in the diagram above.

The performer has mass 80 kg and is modelled as a particle. Initially the performer is held at rest at the centre of the shaft, 7 m above the floor. The shaft is 52 m deep and 48 m wide. Each elastic cord has natural length 30 m and modulus of elasticity 3150 N.

The performer is released from rest.

Air resistance is modelled as negligible.

(a) Show that the speed of the performer when the cords first become slack is $25\,\text{m}\,\text{s}^{-1}$ correct to two significant figures. **[6]**

(b) Determine whether the performer is moving up or down when the cords become taut again. **[5]**

Solution

(a) The performer is initially at the centre of the shaft, so the horizontal distance to each fixing point is

$$\frac{48}{2} = 24\,\text{m}$$

and the vertical distance below the rim is

$$52 - 7 = 45\,\text{m}$$

Hence the initial length of each cord is

$$\sqrt{24^2 + 45^2} = \sqrt{2601} = 51\,\text{m}$$

So the initial extension of each cord is

$$51 - 30 = 21\,\text{m}$$

When the cords first become slack, each cord has its natural length, 30 m. If the performer is then h m above the floor, the vertical distance below the rim is $52 - h$, so

$$24^2 + (52 - h)^2 = 30^2$$

$$(52 - h)^2 = 324$$

The performer is still below the rim at this instant, so

$$52 - h = 18$$

so

$$h = 34$$

Therefore the performer has risen

$$34 - 7 = 27\,\text{m}$$

The elastic potential energy is

$$EPE = \frac{\lambda x^2}{2l}$$

where x is the extension and l the natural length.

The work–energy principle is

$$KE_0 + GPE_0 + EPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1 + EPE_1$$

where state 0 is the moment of release and state 1 the moment the cords first become slack. No other force does work, so $\sum W_{\text{other}} = 0$.

Take the initial position as the zero level for gravitational potential energy. Then

$$0 + 0 + 2\left(\frac{3150 \times 21^2}{2 \times 30}\right) + 0 = \frac{1}{2}(80)v^2 + 80(9.8)(27) + 0$$

$$46\,305 = 40v^2 + 21\,168$$

$$25\,137 = 40v^2$$

$$v^2 = 628.425$$

So

$$v = \sqrt{628.425} \approx 25.1\,\text{m}\,\text{s}^{-1}$$

Hence the speed when the cords first become slack is

$$25\,\text{m}\,\text{s}^{-1} \qquad (2\text{ s.f.})$$

(b) From part (a), when the cords first become slack the performer is 34 m above the floor and has

$$v^2 = 628.425$$

While the cords are slack, there is no elastic potential energy, and no other force does work. Take the position where the cords first become slack as the zero level for gravitational potential energy. At the highest point, H m above the floor, the speed is 0, so the work–energy principle gives

$$\frac{1}{2}(80)v^2 + 0 = 0 + 80(9.8)(H - 34)$$

$$40v^2 = 784(H - 34)$$

Using $v^2 = 628.425$:

$$H = 34 + \frac{628.425}{19.6} \approx 66.1$$

So the greatest height reached is 66.1 m above the floor, which is

$$66.1 - 52 = 14.1\,\text{m}$$

above the rim.

Now suppose the performer is y m above the rim. Then the length of each cord is

$$\sqrt{24^2 + y^2}$$

For the cords to become taut again, this length must be 30 m, so

$$24^2 + y^2 = 30^2$$

$$y^2 = 324$$

$$y = 18$$

So the performer would need to be 18 m above the rim for the cords to become taut on the upward journey. But the maximum height above the rim is only 14.1 m, so this does not happen.

Therefore the cords become taut again only after the performer has turned and started descending.

So the performer is moving down when the cords become taut again.